

PERINBAN PARAMESHWARAN

SENIOR DATA ENGINEER
perinban.github.io/myPortfolio



Profile

Data Engineer with 6+ years of experience building production-grade data infrastructure and a Master's in Data Science. Skilled in designing scalable enterprise data pipelines using Ab Initio (GDE, Control Center, Conduct IT, Express IT, TRW), Oracle SQL, and Shell scripting. Experienced in integrating heterogeneous data sources, automated workflows, and reproducible pipelines using Python, PostgreSQL, Airflow, and AWS. Currently building scientific data infrastructure at UFZ Leipzig. Developer of MetaXuda — an open-source GPU runtime for Apple Silicon with 5K+ downloads. Open-source contributor to llama.cpp

Professional Skills

Ab initio	★	★	★	★	★
Unix Scripting	★	★	★	★	★
SQL	★	★	★	★	★
Conduct IT	★	★	★	★	★
Metadata Hub	★	★	★	★	★
One Automation	★	★	★	★	★
Control Center	★	★	★	★	★
Docker	★	★	★	★	★
Jira	★	★	★	★	★
Python	★	★	★	★	★
Rust	★	★	★	★	★
ML	★	★	★	★	★

Career History

Research Assistant
UFZ

Mar 2026 - Present
Leipzig, Germany

Built a scalable scientific data infrastructure for automated research paper analysis supporting systematic literature reviews:

- **Multi-Source Data Collection:** Collected academic data from multiple different sources (OpenAlex; Google Scholar; Semantic Scholar; Europe PMC; CORE), utilized proxy rotation along with multi-worker architectures.
- **Reproducible Pipeline Architecture:** Implemented end-to-end workflows as a reproducible pipeline from collection to enrichment, screening to structured outputs.
- **Automated PDF Extraction Pipelines:** Utilized OpenCV, pdfPlumber and Playwright to automate the data extraction from scientific figures.
- **PRISMA Relevance Classification:** Created a system to automatically score/ classify relevant articles for systematic review pipelines.
- **Knowledge Base Enrichment:** Developed a structured knowledge base with geospatial enrichments.

Technologies: Python, asyncio, Selenium, spaCy, PostgreSQL, Pandas, Docker, OpenCV, pdfPlumber

Independent Data Engineer

Mar 2024 - Feb 2026
Berlin, Germany

- **Data Engineering:** Built and optimized large-scale enterprise data pipeline automation in Python, Apache Airflow, Apache Kafka, and Hadoop.
- **Cloud & APIs:** Designed AWS architectures and RESTful APIs with ML models for forecasting and anomaly detection

Technologies: Python, Rust, AWS, Airflow, Kafka, PostgreSQL, Pandas, scikit-learn, Git, Docker

Technical Showcase : [Tableau Dashboard](#) | [GitHub](#)

Achievements

Full Achievements and Award
Certificates - [View PDF](#)

Patents & IP

GUS-101 — GPU-accelerated tensor computation method; German utility model (in preparation, 2026), supported by University of Europe for Applied Sciences.

Letter of

Recommendation

- LTIMindtree Manager - [View PDF](#)
- Prof. Dr. Talha Ali Khan (UE Germany) - [View PDF](#)

Education

Mar 2024 - Feb 2026

University of Europe for Applied Sciences, Potsdam

M.Sc. Data Science

Oct 2014 - May 2018

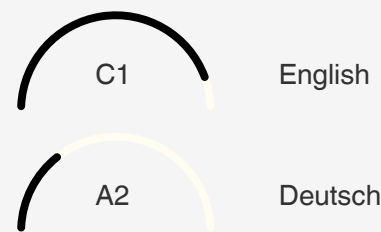
KCG College of Technology, Chennai

B.Tech Information technology

Certifications

- Microsoft Certified: Azure Fundamentals - AZ-900
- AWS Academy Graduate: Cloud Foundations
- KNIME L1 Certification
- Pluralsight Certificates: JavaScript, Python, Tableau

Languages



+49 162 6575372
p.perinban@gmail.com
Plönzeile 27, 12459 Berlin
[linkedin.com/in/perinban-parameshwaran](#)

Senior Data Engineer
LTIMindtree
Module Lead
DATA-FMB-CTO21294-LTI
Oct 2021 - Feb 2024
Chennai, India

Automated regulatory reports for UK PRA branch returns & Turkey reporting; created scalable automation with optimized ETL/ELT:

- **Automation Architect:** Designed scalable Ab Initio ELT workflows that automated UK PRA Branch Returns and Turkey regulatory reporting in addition to batch processes and reconciliation reporting.
- **Performance Optimizer:** Automated and streamlined ETL/ELT workflows and removed process bottlenecks to reduce report generation time by 40%.
- **Quality Guardian:** Achieved a 99.9% level of data quality through QA controls, resolving discrepancies and refining business logic.
- **Technical Lead:** Directed and delivered on the UK PRA regulatory reporting project, provided technical guidance and delivery oversight.
- **Leadership Mentor:** Mentored team members; promoted teamwork among project team members.

Roles and Responsibilities: [Job Duties and Skills](#)
Technologies : Ab initio, Oracle SQL, Korn Shell Scripting, Citi Insight, Reginsight, Control Center, Conduct IT, Express IT (Helix), Query IT, Metadata Hub, RLM, Jira, Jenkins, Git
Associate Consultant
Atos Syntel
FedEx HRT Ab initio Dev Support
May 2018 - Oct 2021
Chennai, India

Developed end-to-end ETL Solutions to integrate global Payroll/HR systems seamlessly:

- **DataHub Integration:** Developed over 35+ Data Interfaces such as SAP SuccessFactors, Vanguard, Ceridian and others to manage employee information, Job History and Organizational Structures.
- **Payroll Automation:** Designed ETL Logic to convert swipe in / swipe out logs to payroll Metrics (Overtime, Weekend Shifts, Bonuses) as well as created Real Time Reconciliation Pipelines (every 2 hours).
- **Cost Savings:** Developed Reconciliation Solutions that eliminated fraudulently reported discrepancies, enhanced Trust and Reduced Operational Costs.
- **Testing & QA:** Automate Test Results Integration with Business Requirements and decreased Manual Testing by 60%.
- **Project Methodologies:** Delivered Projects using Both Agile and Waterfall methodologies across multiple Teams.

Technologies: Ab initio, Oracle SQL, Korn Shell Scripting, One Automation, HP ALM, AGM, Version One, Conduct IT, Metadata Hub, TRW, Azure